



Anisotropic Yield Criterion of Hill

Course: Composite Material Damage Mechanics

Presenter: Rouzbeh Pourjafarian

Instructor: Dr. Hosseini Kordkheili

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Why an Anisotropic Theory?

1 Introduction and Motivation

- **Isotropic plasticity is well established:** the Huber–Mises yield criterion and the Lévy–Mises flow relations describe metals such as copper, aluminium and nickel that deform homogeneously (Hill, 1948).
- **But forming changes the material:** rolling, drawing and extrusion gradually develop a *preferred orientation* of crystal planes and a fibrous texture along the direction of most severe strain.
- **Consequence:** the metal becomes anisotropic. After severe cold rolling the yield stress can be noticeably higher across the fibre than along it, and ductility is even more directional.
- **Engineering symptom:** “earing” of cups deep-drawn from rolled sheet, and direction-dependent necking.



Goal and Target Equation

1 Introduction and Motivation

- **Objective (Hill, 1948):** a macroscopic theory for the yielding and plastic flow of a metal made anisotropic by preferred orientation.
- **The equation we will obtain and prove** — Hill's yield criterion, his Eq. (5):

$$\begin{aligned} 2f(\sigma_{ij}) = & F(\sigma_{yy} - \sigma_{zz})^2 + G(\sigma_{zz} - \sigma_{xx})^2 + H(\sigma_{xx} - \sigma_{yy})^2 \\ & + 2L\tau_{yz}^2 + 2M\tau_{zx}^2 + 2N\tau_{xy}^2 = 1 \end{aligned} \quad (\star)$$

- **Six constants** F, G, H, L, M, N specify the current state of anisotropy, referred to the principal axes of anisotropy x, y, z .
- **Three questions for this talk:** *Why* this equation? *How* is it obtained? *How* do we prove it is the right one (and that it contains the isotropic theory)?



Roadmap of the Chosen Path

1 Introduction and Motivation





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Elastic Strain Energy (Huber, 1904)

2 The Isotropic Baseline (Huber–Mises)

- For proportional loading from the unstressed state, the stored energy per unit volume is half the conjugate stress–strain products:

$$F = \frac{1}{2} (\sigma_1 \varepsilon_1 + \sigma_2 \varepsilon_2 + \sigma_3 \varepsilon_3). \quad (1)$$

- Isotropic Hooke's law, $\varepsilon_i = \frac{1}{E} [\sigma_i - \mu(\sigma_j + \sigma_k)]$, gives after substitution and collecting terms:

$$F = \frac{1}{2E} [\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)]. \quad (2)$$

- This is the basis of everything that follows: the total elastic energy in terms of the principal stresses and the elastic constants E, μ .



Volumetric / Distortional Split

2 The Isotropic Baseline (Huber–Mises)

- **Physical fact:** pure hydrostatic pressure, however large, does not bring a ductile metal to yield. Only the energy of *shape change* can drive plastic flow.
- Split $F = F_{\text{vol}} + F_{\text{dist}}$ about the mean stress $p = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$:

$$F_{\text{vol}} = \frac{1 - 2\mu}{6E} (\sigma_1 + \sigma_2 + \sigma_3)^2. \quad (3)$$

- Subtracting F_{vol} from F and using $(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2(P - Q)$, with $P = \sum \sigma_i^2$, $Q = \sum \sigma_i \sigma_j$:

$$F_{\text{dist}} = \frac{1 + \mu}{6E} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]. \quad (4)$$

- **Checks:** hydrostatic state $\Rightarrow F_{\text{dist}} = 0$; isochoric state $\Rightarrow F_{\text{vol}} = 0$.



The Second Invariant J_2 (von Mises, 1928)

2 The Isotropic Baseline (Huber–Mises)

- Deviatoric stress (mean stress removed from the diagonal):

$$S_{ij} = \sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij}, \quad S_{kk} = 0. \quad (5)$$

- Its second invariant, evaluated on principal axes:

$$J_2 = \frac{1}{2} S_{ij} S_{ij} = \frac{1}{3} (P - Q). \quad (6)$$

- Hence the distortional energy is proportional to a frame-independent invariant:

$$F_{\text{dist}} = \frac{1 + \mu}{3E} (P - Q) = \frac{1 + \mu}{E} J_2. \quad (7)$$

- **Key point:** for an isotropic body the material dependence of F_{dist} sits entirely in the scalar factor $(1 + \mu)/E$; the *stress part* is the pure invariant J_2 . (Remember this; it fails under anisotropy.)



Yield Hypothesis and Uniaxial Tensile

2 The Isotropic Baseline (Huber–Mises)

- **Hypothesis:** yielding begins when F_{dist} reaches the value it attains at the uniaxial tensile yield point of the same material:

$$F_{\text{dist}} = F_{\text{dist}}^{\text{uniaxial}}. \quad (8)$$

- **Calibration:** in uniaxial tension $\sigma_1 = Y$, $\sigma_2 = \sigma_3 = 0$, so from Eq. (4):

$$F_{\text{dist}}^{\text{uniaxial}} = \frac{1 + \mu}{6E} (Y^2 + 0 + Y^2) = \frac{1 + \mu}{3E} Y^2. \quad (9)$$

- Equating (4) and (9), the common factor $(1 + \mu)/E$ cancels; the elastic constants disappear:

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2Y^2. \quad (10)$$

- Eq. (10) is the Huber–Mises criterion in principal stresses; it depends only on the single constant Y .



Huber–Mises in General Components

2 The Isotropic Baseline (Huber–Mises)

- Using stress invariants $I_1 = \sigma_{kk}$ and I_2 , the left side of (10) equals $2I_1^2 - 6I_2$, which is frame-independent. Evaluated in the general x, y, z frame it expands to:

$$(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\tau_{yz}^2 + \tau_{zx}^2 + \tau_{xy}^2) = 2Y^2. \quad (11)$$

- This is the Huber–Mises criterion in general stress components; the form Hill writes as his Eq. (3) and the isotropic limit his anisotropic theory *must* reproduce.



Setting Up Eq. (11) via Invariants

2 The Isotropic Baseline (Huber–Mises)

- **Idea:** the left side of (10) is a *scalar invariant*; independent of axes. Rewrite it via I_1, I_2 (same in any frame), then re-express I_1, I_2 in general axes; the result is then automatically valid for any axes.
- **Recall** $Q = I_2$ and $I_1^2 = P + 2Q \Rightarrow P = I_1^2 - 2I_2$. So the left side of (10), $2(P - Q)$, becomes

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2(P - Q) = 2I_1^2 - 6I_2,$$

true in *any* axes since I_1, I_2 are invariant.

- **General-axes forms.** I_1 is just the trace:

$$I_1 = \sigma_{xx} + \sigma_{yy} + \sigma_{zz}.$$

I_2 is the sum of the principal 2×2 minors of the stress matrix:

$$I_2 = \sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx} - (\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2).$$

This is where shear stresses enter I_2 ; with a *minus* sign.



Assembling Eq. (11) from I_1, I_2

2 The Isotropic Baseline (Huber–Mises)

- **Expand I_1^2 :**

$$I_1^2 = \sigma_{xx}^2 + \sigma_{yy}^2 + \sigma_{zz}^2 + 2(\sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx}).$$

- **Substitute into $2I_1^2 - 6I_2$** and collect the normal-stress cross terms (+4 from $2I_1^2$, minus 6 from $6I_2$, net -2), while the shear terms in $6I_2$ flip sign to $+6$:

$$2I_1^2 - 6I_2 = 2(\sigma_{xx}^2 + \sigma_{yy}^2 + \sigma_{zz}^2) - 2(\sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx}) + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2).$$

- **Recognize the normal-stress part** as a sum of squared differences (expand each square and sum to check):

$$\begin{aligned} & 2(\sigma_{xx}^2 + \sigma_{yy}^2 + \sigma_{zz}^2) - 2(\sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx}) \\ &= (\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2. \end{aligned}$$



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The Meaning of $f(\sigma_{ij}) = \text{const}$

3 The Plastic Potential (von Mises, 1928)

- **Geometric meaning:** $f = \text{const}$ is a *level surface*, the yield surface, in stress space, separating safe ($f < \text{const}$) from yielding ($f = \text{const}$) states.
- **The constant is the threshold itself** a material property, *measured* (not derived) by a simple calibration test: uniaxial tension gives $F_{\text{dist}} = F_{\text{dist}}^{\text{uniaxial}}$ (Eq. 8); Hill's six tests, Eqs. (17)–(19), play the same role.



The Plastic Potential and Flow Rule

3 The Plastic Potential (von Mises, 1928)

- Let the yield criterion be $f(\sigma_{ij}) = \text{const}$ (Hill Eq. 1). Von Mises' idea: the plastic strain increment is directed along the gradient of f in stress space,

$$d\varepsilon_{ij} = d\lambda \frac{\partial f}{\partial \sigma_{ij}}, \quad d\lambda > 0. \quad (12)$$

- $d\lambda$ is a positive scalar set by the loading history, not by the criterion. The increment is *normal* to the surface $f = \text{const}$ (normality rule).
- Differentiation convention:** the nine components σ_{ij} are treated as distinct independent variables, with σ_{ij} and σ_{ji} ($i \neq j$) counted separately.
- For an isotropic metal f does not depend on the axes, so (12) is a true tensor relation; for an anisotropic metal the form of f depends on the chosen axes.



Check: the Rule Reproduces Lévy–Mises

3 The Plastic Potential (von Mises, 1928)

- Take f as half of the left side of (11), $f = \frac{1}{2}[(\sigma_{xx} - \sigma_{yy})^2 + \dots] + 3(\tau_{yz}^2 + \tau_{zx}^2 + \tau_{xy}^2)$.
- Normal component (only the 1st and 3rd squared differences contain σ_{xx}):

$$\frac{\partial f}{\partial \sigma_{xx}} = (\sigma_{xx} - \sigma_{yy}) + (\sigma_{xx} - \sigma_{zz}) = 2\sigma_{xx} - \sigma_{yy} - \sigma_{zz}. \quad (13)$$

- Shear component (only $3\tau_{yz}^2$ contributes), with engineering shear $\gamma_{yz} = 2\varepsilon_{yz}$ conjugate to τ_{yz} :

$$d\varepsilon_{xx} = d\lambda(2\sigma_{xx} - \sigma_{yy} - \sigma_{zz}), \quad d\varepsilon_{yz} = 3d\lambda\tau_{yz}. \quad (14)$$

- These are exactly the Lévy–Mises relations (Hill Eq. 4): the plastic-potential idea passes its isotropic test.



Why the Potential Works: Stationary Work

3 The Plastic Potential (von Mises, 1928)

- Von Mises gave the rule a physical meaning. For a fixed plastic strain increment $d\varepsilon_{ij}$, vary the stress subject only to $f(\sigma_{ij}) = \text{const}$.
- The work $\sigma_{ij} d\varepsilon_{ij}$ is stationary at the actual stress. With a Lagrange multiplier $d\lambda$:

$$\delta(\sigma_{ij} d\varepsilon_{ij} - f d\lambda) = 0 \implies d\varepsilon_{ij} = d\lambda \frac{\partial f}{\partial \sigma_{ij}}. \quad (15)$$

- For the Huber–Mises f this stationary value is a *maximum* (maximum-work principle). For Hill's criterion it remains a maximum except possibly when one of F, G, H is negative.
- This interpretation will also link *hydrostatic independence* to *zero plastic volume change* (used later).



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Three Assumptions

4 The Anisotropic Criterion of Hill

- **(A1) Orthotropy:** three mutually orthogonal planes of symmetry at every point. Their intersections are the *principal axes of anisotropy*, taken as x, y, z . (They may rotate from point to point and with continued deformation.)
- **(A2) No Bauschinger effect:** tensile and compressive yield stresses are equal along each axis. The yield function is therefore *even* in the stresses; the simplest such function is a *homogeneous quadratic*.
- **(A3) Hydrostatic independence:** superimposing a pressure p does not affect yielding, $f(\sigma_{ij} + p \delta_{ij}) = f(\sigma_{ij})$.



Why *Postulate*, Not Derive from Energy?

4 The Anisotropic Criterion of Hill

Tempting idea: by analogy with Hencky's energy reading of Huber–Mises, take f as the anisotropic shear-strain (distortional) energy. Hill rejects this for *two* reasons:

- **Objection 1: hydrostatic coupling.** In an anisotropic *elastic* medium a hydrostatic pressure generally produces shear strain. So the shear-strain energy would change with superposed pressure, implying pressure affects yielding, contradicting (A3).
- **Objection 2: elastic constants intrude.** Unlike the isotropic case (where the stress part of F_{dist} is the pure invariant J_2 , recall Eq. 7), the anisotropic shear-strain energy is *not* independent of the elastic constants apart from a common factor. They would then appear in the plastic strain increment (12), which is not physically plausible.

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Building the Form from Symmetry

4 The Anisotropic Criterion of Hill

Start from the isotropic criterion (11) read as a quadratic, and let symmetry decide which terms survive:

- **(A2) even** $\Rightarrow$ homogeneous quadratic; terms linear in any shear stress are rejected.
- **(A1) orthotropy** (invariance under reflection in each coordinate plane) $\Rightarrow$ admit only terms *even in every shear stress*: this deletes every normal $\times$ shear product and every product of two *different* shears, leaving only the squares $\tau_{yz}^2, \tau_{zx}^2, \tau_{xy}^2$ plus a quadratic in the normal stresses.
- **(A3) hydrostatic independence** $\Rightarrow$ the normal stresses enter only through their *differences* $\sigma_{ii} - \sigma_{jj}$ (a difference is unchanged when the same p is added to each).

The most general quadratic meeting (A1)–(A3) is a weighted sum of the three squared normal differences and the three squared shears — the isotropic form (11) with its coefficients *set free*.



The Anisotropic Criterion (Hill, Eq. 5)

4 The Anisotropic Criterion of Hill

- Writing the free coefficients F, G, H (normal differences) and $2L, 2M, 2N$ (shear squares), and normalising the right side to unity so that yield $\Leftrightarrow 2f = 1$:

$$\begin{aligned} 2f(\sigma_{ij}) = & F(\sigma_{yy} - \sigma_{zz})^2 + G(\sigma_{zz} - \sigma_{xx})^2 + H(\sigma_{xx} - \sigma_{yy})^2 \\ & + 2L \tau_{yz}^2 + 2M \tau_{zx}^2 + 2N \tau_{xy}^2 = 1 \end{aligned} \quad (16)$$

- F, G, H, L, M, N characterise the *current* state of anisotropy. The form holds only when the axes are the principal axes of anisotropy; other axes change the form via stress transformation.
- By construction it stands to the isotropic criterion (11) as “six free coefficients” stands to “those coefficients fixed equal.”



F, G, H from Tension Tests

4 The Anisotropic Criterion of Hill

- Uniaxial tension $\sigma_{xx} = X$ (others zero) in (16): $(\sigma_{zz} - \sigma_{xx})^2 = (\sigma_{xx} - \sigma_{yy})^2 = X^2$,
so

$$\frac{1}{X^2} = G + H, \quad \frac{1}{Y^2} = H + F, \quad \frac{1}{Z^2} = F + G. \quad (17)$$

- Adding the three: $F + G + H = \frac{1}{2}(\frac{1}{X^2} + \frac{1}{Y^2} + \frac{1}{Z^2})$. Subtracting each in turn:

$$2F = \frac{1}{Y^2} + \frac{1}{Z^2} - \frac{1}{X^2}, \quad 2G = \frac{1}{Z^2} + \frac{1}{X^2} - \frac{1}{Y^2}, \quad 2H = \frac{1}{X^2} + \frac{1}{Y^2} - \frac{1}{Z^2}. \quad (18)$$

- At most one of F, G, H can be negative (it would require large differences among X, Y, Z , unlikely in practice).



Calibrating L, M, N from Shear Tests

4 The Anisotropic Criterion of Hill

- A pure shear $\tau_{yz} = R$ at yield leaves only $2L \tau_{yz}^2$ in (16):

$$2L = \frac{1}{R^2}, \quad 2M = \frac{1}{S^2}, \quad 2N = \frac{1}{T^2}, \quad (19)$$

where R, S, T are the shear yield stresses about the principal axes.

- Since $R^2, S^2, T^2 > 0$, the constants L, M, N are *essentially positive*.
- **State of anisotropy** is fully described by six yield stresses X, Y, Z, R, S, T , which depend on the strain history and so evolve with deformation.



Associated Flow Rule and Zero Volume Change

4 The Anisotropic Criterion of Hill

- Apply the plastic potential (12) to the criterion (16). The normal increments are

$$\begin{aligned} d\varepsilon_{xx} &= d\lambda [H(\sigma_{xx} - \sigma_{yy}) + G(\sigma_{xx} - \sigma_{zz})] \quad (\text{and cyclic in } x, y, z), \\ d\varepsilon_{yz} &= d\lambda L \tau_{yz}, \quad d\varepsilon_{zx} = d\lambda M \tau_{zx}, \quad d\varepsilon_{xy} = d\lambda N \tau_{xy}. \end{aligned} \quad (20)$$

- **Proof of isochoric flow:** sum the three normal increments. The H pair gives $H(\sigma_{xx} - \sigma_{yy}) + H(\sigma_{yy} - \sigma_{xx}) = 0$; likewise the G and F pairs cancel, so

$$d\varepsilon_{xx} + d\varepsilon_{yy} + d\varepsilon_{zz} = 0. \quad (21)$$

- This zero plastic volume change is *forced by* assumption (A3): via the stationary-work view, a superposed pressure does no extra work when the volume change is zero — so hydrostatic independence and isochoric flow imply each other for *any* yield criterion.
- (Shear increments use the engineering-shear convention $d\gamma_{yz} = 2 d\varepsilon_{yz}$, as in the isotropic check Eq. 14.)



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Proof: Reduction to Isotropy

5 Consistency and Closure

- Isotropy means equal tensile yield stresses $X = Y = Z$ and equal shear yield stresses $R = S = T = Y/\sqrt{3}$ (the Huber–Mises value, since $6T^2 = 2Y^2$).
- Tension: from (18), $F = G = H = \frac{1}{2Y^2}$. Shear: from (19), $2L = \frac{1}{R^2} = \frac{3}{Y^2}$, so $L = M = N = \frac{3}{2Y^2} = 3F$.
- Substitute into (16) and multiply by $2Y^2$:

$$(\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + (\sigma_{xx} - \sigma_{yy})^2 + 6(\tau_{yz}^2 + \tau_{zx}^2 + \tau_{xy}^2) = 2Y^2. \quad (22)$$

- Eq. (22) is exactly the Huber–Mises criterion (11). The anisotropic criterion contains the



Special Case and Summary of the Argument

5 Consistency and Closure

- **Transverse isotropy** about z (rotational symmetry) is recovered by requiring f invariant for arbitrary x, y axes: $F = G$ and $N = F + 2H$ a further consistency check on the framework.
- **The logical chain:**
 1. Energy $\rightarrow$ isotropic Huber–Mises (Eqs. 1–11).
 2. Plastic potential $\rightarrow$ flow rule, verified on the isotropic case (Eqs. 12–15).
 3. Reason: energy route *fails* for anisotropy (two objections) $\Rightarrow$ postulate f .
 4. Symmetry assumptions (A1)–(A3) $\Rightarrow$ the unique quadratic form, Eq. (16).
 5. Six tests calibrate $F, \dots, N$; flow rule gives isochoric plasticity; isotropy is recovered.
- **Why this equation:** it is the simplest stress function consistent with orthotropic symmetry, no Bauschinger effect, and pressure-independent yielding and it reduces correctly to the trusted isotropic theory.



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Q&A

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Your feedback will be highly appreciated!